

- 4 The photograph shows a species of harvest mouse native to countries in Asia.



(Source: © Mark Bridger/Shutterstock)

- (a) The body of this species of harvest mouse is 55 mm to 75 mm long and its tail is 50 mm to 75 mm long.

The body of the Indian flat-haired mouse is 10 cm to 12 cm long and its tail is 7 cm to 8 cm long.

Calculate the percentage difference in **minimum** total length of the Indian flat-haired mouse compared with the **minimum** total length of the harvest mouse.

Give your answer to **two** significant figures.

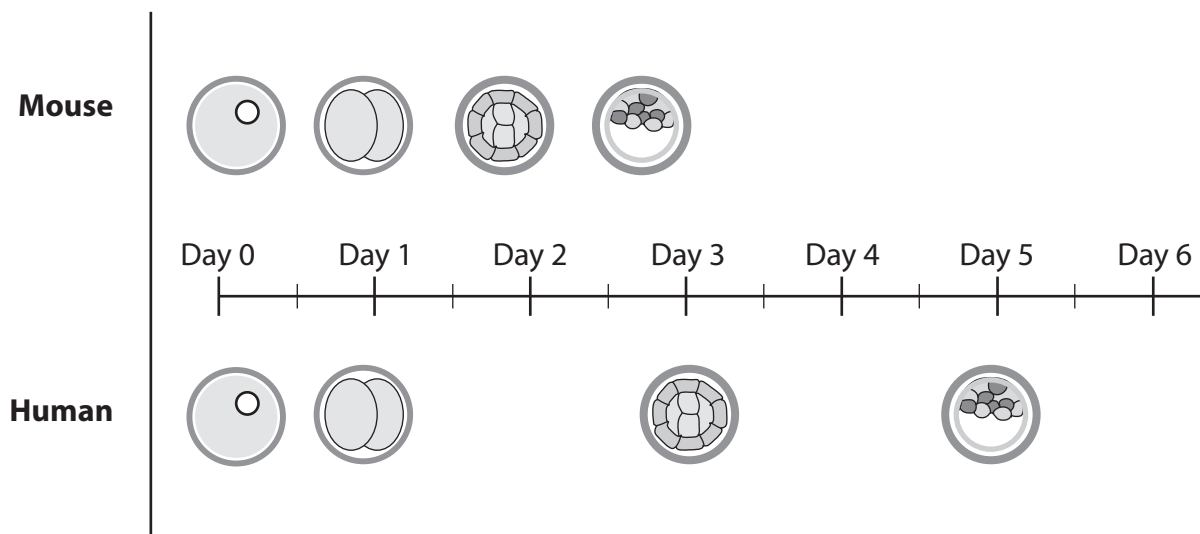
(2)

Answer %



- (b) The diagram shows the timeline for the structures formed after fertilisation in a mouse and in a human.

A fertilised egg cell is called a zygote.



- (i) As mouse egg cells develop they accumulate small spherical lipid droplets. These lipid droplets have a mean radius of $88 \mu\text{m}$. The volume of a sphere (V) is calculated using the formula

$$V = \frac{4}{3} \pi r^3$$

Which of the following gives the mean volume of a lipid droplet in mm^3 ?

(1)

- ☐ A 2.85×10^{-6}
- ☐ B 2.85×10^{-3}
- ☐ C 2.85×10^3
- ☐ D 2.85×10^6

- (ii) Draw a circle around the part of the diagram that is a mouse morula. Label the circle 'morula'.

(1)



(iii) Describe the events that occur after a sperm cell enters an egg cell, until a zygote is formed.

(3)

DO NOT WRITE IN THIS AREA

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(c) Analysis of some mouse egg cells showed that the lipid droplets were located within 10 nm of mitochondria.

Suggest why mitochondria were located within 10 nm of the lipid droplets.

(3)

(Total for Question 4 = 10 marks)

